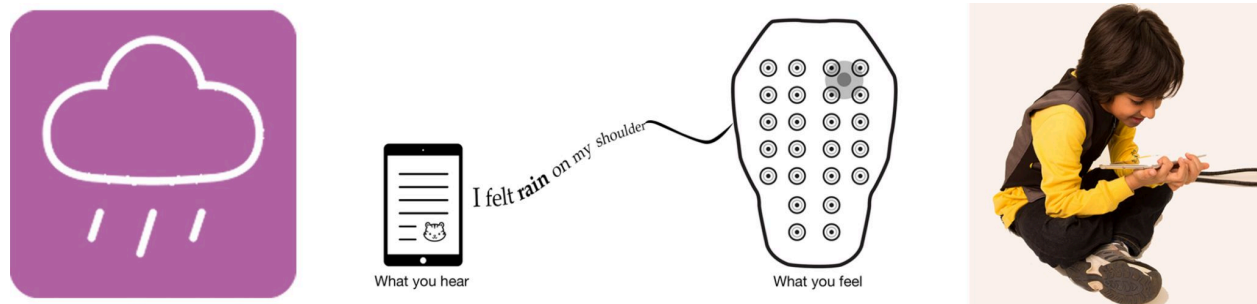


Semantic based Parametric Haptics for communication, storytelling, and social networking

- Israr, A., Zhao, S., Schwalje, K., Klatzky, R., & Lehman, J. (2014). Feel effects: enriching storytelling with haptic feedback. *ACM Transactions on Applied Perception (TAP)*, 11(3), 1-17. [\[Link\]](#)

Despite a long history of use in communication, haptic feedback is a relatively new addition to the toolbox of special effects. Unlike artists who use sound or vision, haptic designers cannot simply access libraries of effects that map cleanly to media content, and they lack even guiding principles for creating such effects. In this project, we make progress toward both capabilities: we generate a foundational library of usable haptic vocabulary and do so with a methodology that allows ongoing additions to the library in a principled and effective way. We define a **feel effect** as an explicit pairing between a meaningful linguistic phrase and a rendered haptic pattern. Our initial experiment demonstrates that users who have only their intrinsic language capacities, and no haptic expertise, can generate a core set of feel effects that lend themselves via semantic inference to the design of additional effects. Our experiments demonstrate a unique and systematic approach to designing a vocabulary of haptic sensations that are related in both the semantic and parametric spaces.



FeelCraft is a media plugin that monitors events and states in the media and associates them with expressive tactile content using a library of feel effects (FEs). A feel effect (FE) is a user-defined haptic pattern that, by virtue of its connection to a meaningful event, generates dynamic and expressive effects on the user's body. We compiled a library of more than fifty FEs associated with common events in games, movies, storybooks, etc., and used them in a sandbox-type gaming platform. The FeelCraft plugin allows a game designer to quickly generate haptic effects, associate them to events in the game, play them back for testing, save them and/or broadcast them to other users to feel the same haptic experience. Our demonstration shows an interactive procedure for authoring haptic media content using the FE library, playing it back during interactions in the game, and broadcasting it to a group of guests.

- Zhao, S., Schneider, O., Klatzky, R., Lehman, J., & Israr, A. (2014, October). FeelCraft: crafting tactile experiences for media using a feel effect library. In *Adjunct Proceedings of the 27th Annual ACM Symposium on User Interface Software and Technology* (pp. 51-52). [\[Link\]](#)

- Schneider, O., Zhao, S., & Israr, A. (2015). FeelCraft: User-crafted tactile content. In *Haptic Interaction: Perception, Devices and Applications* (pp. 253-259). Tokyo: Springer Japan. [\[Link\]](#)



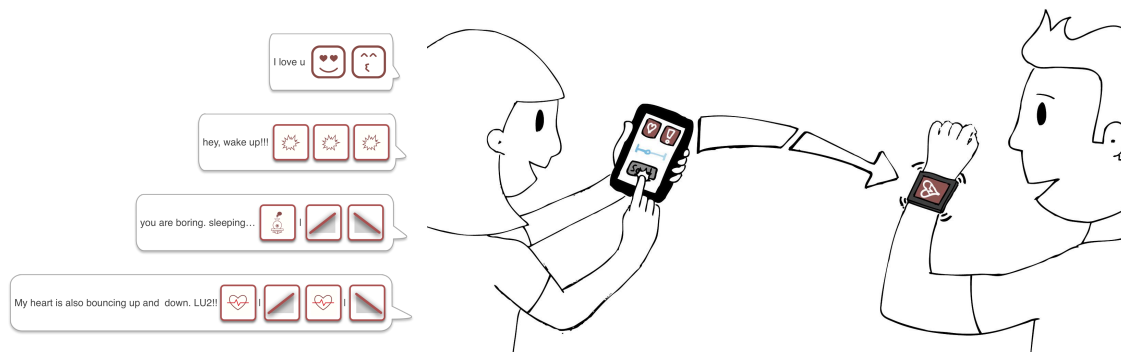
@  FeelCraft: Crafting Tactile Experiences for Media using a Feel Effect Library

- Yannier, N., Israr, A., Lehman, J. F., & Klatzky, R. L. (2015, April). FeelSleeve: Haptic feedback to enhance early reading. In *Proceedings of the 33rd Annual ACM Conference on Human Factors in Computing Systems* (pp. 1015-1024). [\[Link\]](#)

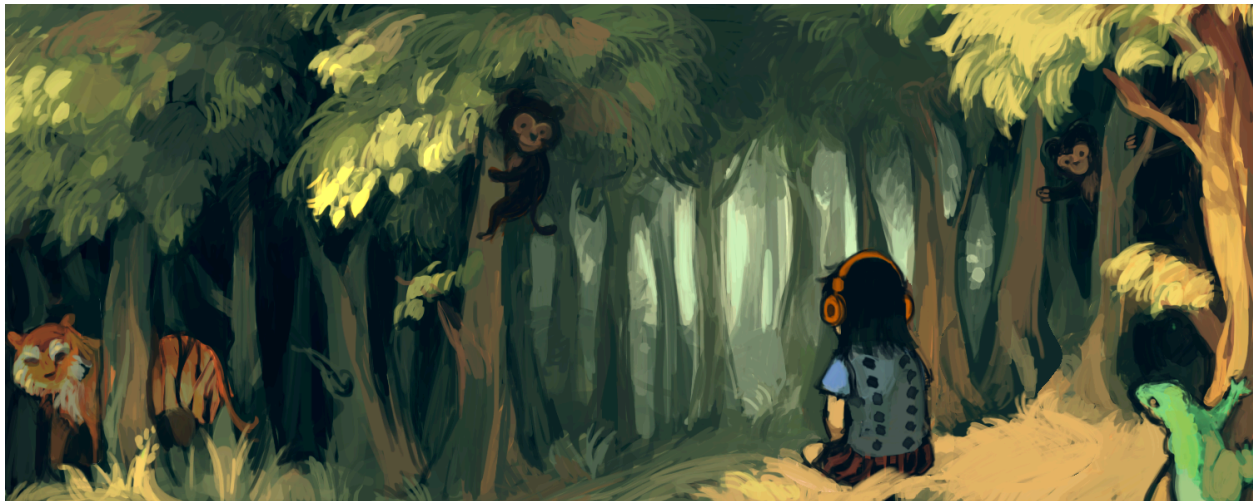


@<https://www.youtube.com/watch?v=WXabCF19IFU>

- Israr, A., Zhao, S., & Schneider, O. (2015, April). Exploring embedded haptics for social networking and interactions. In *Proceedings of the 33rd Annual ACM Conference Extended Abstracts on Human Factors in Computing Systems* (pp. 1899-1904). [\[Link\]](#)



- Zhao, S., Lehman, J., Israr, A., & Klatzky, R. (2015, June). Using haptic inputs to enrich story listening for young children. In *Proceedings of the 14th international conference on interaction design and children* (pp. 239-242). [\[Link\]](#)



Based on the research, we build a haptic LLM that generates semantically coherent spatial haptic patterns on a grid display.

Haptics GenAI Studio



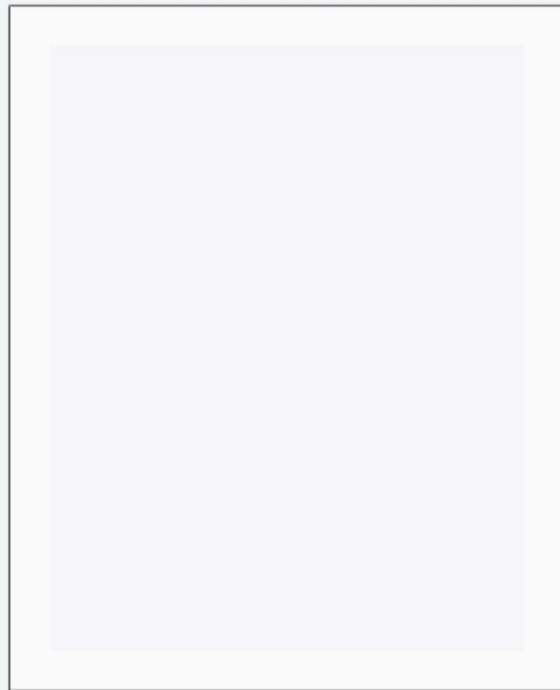
Powered by ViiVAI Labs

Enter Prompt:

Generate Haptic Pattern

Play Haptic Pattern

Ready



@  HapticGenAI demo